Task 1

Use the information sheets to fill in the table.

Feature	Catalyst	Enzyme
Function		
Rate of reaction		
Range of reactions (specificity)		
Structure		
Required temperatures		
How the reaction occurs		

Task 1 – standard information sheets

Function

Enzymes have the same properties as chemical catalysts.

They lower the amount of energy needed to get the reaction started (activation energy), as well as increasing the speed of the reaction once it has started.

Rate of reaction

Enzymes are more effective at speeding up reactions than chemical catalysts e.g. the decomposition of hydrogen peroxide (H_2O_2), is accelerated 3 x 10^4 times in the presence of ferric ions which act as a catalyst; but it is accelerated 1 x 10^8 times in the presence of the enzyme catalase.

Range of reactions

Chemical catalysts tend to work on a wider range of reactions, whereas enzymes only bind to one specific substrate.

Structure

Like catalysts, enzymes are not permanently altered by the reaction and can be re-used.

Almost all enzymes consist of proteins which are large molecules with a complex structure. Many catalysts are ions, elements or quite simple inorganic compounds.

Temperature

Because they are produced by living organisms, enzymes work best at relatively low temperatures in contrast with catalysts which generally require high temperatures or pressure to be effective.

How the reaction occurs

Catalysts speed up a reaction by giving it another pathway with a lower activation energy.

Catalysts work by combining with a chemical to make a new compound.

This is unstable and quickly breaks down releasing the products and the catalyst.

This is also how many enzymes work. The substrate combines with the active site of the enzyme and new products are released. This is called the lock and key

theory.

In both cases, the catalyst and the enzyme are both left unchanged at the end of the reaction.

Task 1 - challenge

Use the information sheets to fill in the table.

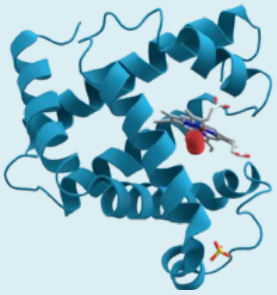
Feature	Catalyst	Enzyme

Task 1 – challenge information sheets

Enzymes have the same properties as chemical catalysts.

They lower the amount of energy needed to get the reaction started (activation energy), as well as increasing the speed of the reaction once it has started.

However, enzymes are more effective at this than chemical catalysts e.g. the decomposition of hydrogen peroxide (H_2O_2), is accelerated 3×10^4 times in the presence of ferric ions which act as a catalyst; but it is accelerated 1×10^8 times in the presence of the enzyme catalase.



Chemical catalysts tend to work on a wide range of reactions, whereas enzymes only bind to one specific substrate.

Like catalysts, enzymes are not permanently altered by the reaction and can be re-used.

Almost all consist of proteins which are large molecules with a complex structure.

Many catalysts are ions, elements or quite simple inorganic compounds.

Because they are produced by living organisms, enzymes work best at relatively low temperatures, in contrast with catalysts which generally require high temperatures or pressure to be effective.

Catalysts speed up a reaction by giving it an alternative pathway which has a lower activation energy.

Enzymes work by the substrate binding to the active site on the enzyme. The active site is an area on the surface of the enzyme that has a specific shape. This means that it will only fit a particular substrate. The substrate fits into the active site and the reaction takes place. The products are released, leaving the enzyme unchanged. This mechanism is called the lock and key theory.

The activity of enzymes can be affected by different factors e.g. pH or temperature.

Task 2 – standard version

Use the information cards to fill in the table.

Substrate	Enzyme	Exocrine tissue which secretes enzyme	Other substances secreted and their effect on conditions for enzyme activity	Products
	amylase			
		gastric pits in stomach		
protein	protease			
				fatty acids and glycerol

Task 2 – challenge version

Use the information cards to fill in the table.

Substrate	Enzyme	Exocrine tissue which secretes enzyme	Other substances secreted and their effect on conditions for enzyme activity	Products

Task 2 – information cards

Salivary glands release amylase which starts to break down carbohydrates, producing glucose.

Gastric pits in the lining of the stomach secrete hydrochloric acid and pepsin which is an enzyme which breaks down proteins into amino acids.

The hydrochloric acid helps to kill microorganisms as well as lowering the pH to allow pepsin to work at its optimum.

After the stomach, food travels to the small intestine.

The enzymes in the small intestine work best in alkaline conditions, so it neutralises the food which is acidic having come from the stomach.

Bile is an alkaline substance produced by the liver and stored in the gall bladder.

It is secreted into the small intestine, where it emulsifies fats.

This is important, because it provides a larger surface area in which the lipases can work.

The pancreas secretes protease, amylase and lipase enzymes into the small intestine.

The simultaneous addition of bile into the small intestine makes sure that the pH is raised to meet the optimum required for these enzymes.

Lipases break down fats into fatty acids and glycerol.

Because fatty acids are formed the pH will be lowered – further production of bile (which is alkaline) will ensure continued action of lipase.

Proteases break down proteins into amino acids.

Amylase breaks down any remaining carbohydrates into glucose.

Teaching notes

Task 1 - starter activity

Before the lesson copy and cut up the statements – there are standard and challenge versions for differentiation. Give each student (or pair of students) a statement and ask them to read it. Ask them to identify a similarity and/or difference between enzymes and catalysts from the statement.

Higher tier

Using the last statements can they draw a diagram to illustrate the lock and key theory? This could be done in pairs initially and then as a class.

Ask students to predict how changing factors such as temperature or substrate concentration may affect enzyme activity and rate of reaction. Can they link this to collision theory and the lock and key theory? Ask them to suggest how they could test their ideas.

Task 2

Differentiate by only giving one section to each student to fill in, then complete on the board together after. OR increase demand by not giving the ‘feature’ column – expect students to fill this in themselves.

Before the lesson copy the statements and give each student (or pair) one statement. Display the blank table on the board and ask the students to fill it in as a class.